

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

ANSWER:

A coin is tossed 10 times:

H H T H H H T H H T

Problem Understanding

Estimate the probability of getting Heads using Maximum Likelihood Estimation (MLE).

Method Selection

For Bernoulli trials,

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Solution Process

- Total tosses = 10
- Number of Heads = 7

$$\hat{p} = \frac{7}{10} = 0.7$$

Accuracy of Calculation

$$\hat{p} = 0.7$$

Interpretation of Results

The estimated probability of getting a Head is **0.7 (70%)**.

2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

ANSWER:

Out of **50 students, 42 passed**.

Problem Understanding

Estimate the probability that a student passes.

Method Selection

MLE:

$$\hat{p} = \frac{\text{Passed}}{\text{Total Students}}$$

Solution Process

$$\hat{p} = \frac{42}{50} = 0.84$$

Accuracy of Calculation

$$\hat{p} = 0.84$$

Interpretation of Results

The estimated probability that a student passes is **0.84 (84%)**.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

ANSWER:

100 bulbs are tested and **8 are defective**.

Problem Understanding

Estimate the probability of a bulb being defective.

Method Selection

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Solution Process

$$\hat{p} = \frac{8}{100} = 0.08$$

Accuracy of Calculation

$$\hat{p} = 0.08$$

Interpretation of Results

The estimated defect probability is **0.08 (8%)**

4. A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

ANSWER:

A die is rolled **60 times** and **6 appears 18 times**.

Problem Understanding

Estimate the probability of rolling a 6.

Method Selection

$$\hat{p} = \frac{\text{Number of sixes}}{\text{Total Rolls}}$$

Solution Process

$$\hat{p} = \frac{18}{60} = 0.3$$

Accuracy of Calculation

$$\boxed{\hat{p} = 0.3}$$

Interpretation of Results

The estimated probability of rolling a 6 is **0.3 (30%)**.

5. Given: Input (x) = 4

Weight (w) = 2

Actual Output =

10 Prediction: y

= w × x

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

ANSWER:

Given:

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Learning Rate = 0.01
- Gradient = 4

Problem Understanding

Find:

1. Predicted Output
2. Error
3. Updated Weight

Method Selection

Prediction:

$$y = w \times x$$

Error:

$$\text{Error} = \text{Actual} - \text{Predicted}$$

Weight Update:

$$w_{\text{new}} = w - \eta \times \text{gradient}$$

where

$$\eta = 0.01$$

Solution Process

Step 1: Prediction

$$y = 2 \times 4 = 8$$

Step 2: Error

$$\text{Error} = 10 - 8 = 2$$

Step 3: Updated Weight

$$\begin{aligned} w_{\text{new}} &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96 \end{aligned}$$

Accuracy of Calculation

Predicted Output = **8**

Error = **2**

Updated Weight = **1.96**

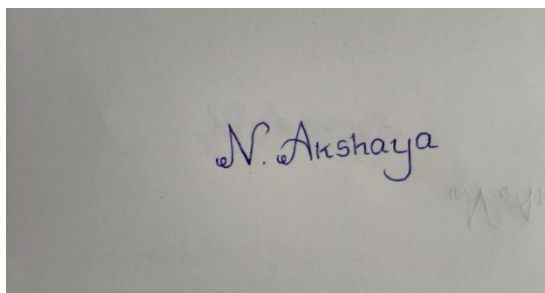
Interpretation of Results

The model predicted **8**, while the actual output was **10**, resulting in an error of **2**. After one gradient descent update, the weight is adjusted from **2** to **1.96**, helping reduce future prediction error.

Code of Conduct:

I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation